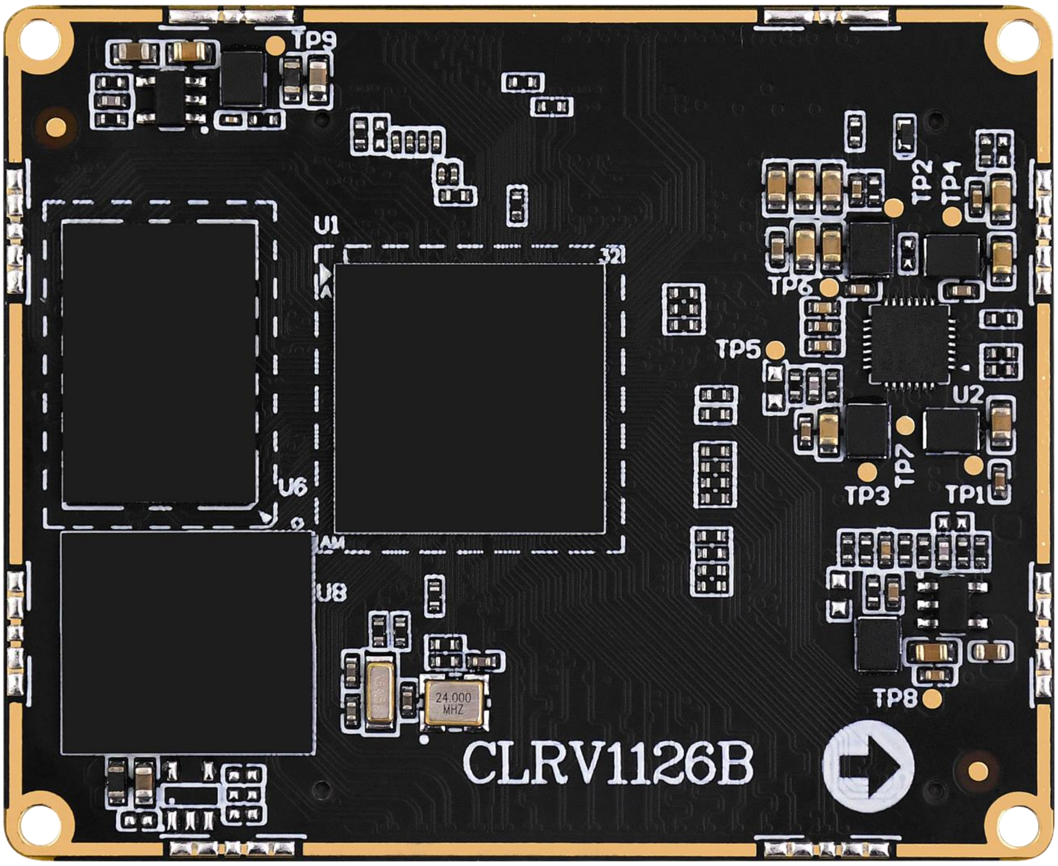
ATK-CLRV1126B

Core Board User Reference Manual

V1.0



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Catalogue

Preface 1
Chapter 1. Research on resources of ATK-CLRV1126B core board..2
    1.1 Overview..2
    1.2 Core Board Pin Order and Exported Interface Signals 2
    1.3 Core Board Pin Multiplexing Resources...3
Chapter 2. Notes on Core Board Usage 5
Chapter 3. Minimum System Design Recommendations..7
    3.1 System Power Supply Design 7
    3.2 Development Board Reset Circuit Design 8
    3.3 Development Board Function Button Design 8
    3.4 UART Debug Interface...10

Preface

Overview

This document mainly introduces the key points and precautions for secondary development and design using the ALIENTEK ATK-CLRV1126B core board, including hardware design and software modification. It is intended to help customers shorten product design cycles, improve design stability, and reduce failure rates.

This document is for reference only. If peripherals are modified for special reasons, the design must strictly comply with high-speed digital circuit design requirements. When developing with the core board, it is recommended to flash the latest factory system image.

Product Model

The core board model corresponding to this document is:

ALIENTEK ATK-CLRV1126B Core Board (hereinafter referred to as the core board)

Target Users

This document is mainly intended for the following engineers:

Hardware development engineers

Driver development engineers

Test engineers

Chapter 1. Research on resources of ATK-CLRV1126B core board

1.1 Overview

The ATK-CLRV1126B Core Board is based on the domestic Rockchip RV1126B/BJ processor, equipped with a quad-core high-performance chip: 4×Cortex-A53 (up to 1.6 GHz), and a 3-TOPS NPU. It supports a 2D Graphics Engine (RGA), 4K hardware codec, and ISP.

The core board measures only 45 mm × 55 mm, and the matching baseboard is 140 mm × 180 mm. Commercial or industrial-grade core boards are available in multiple configurations. It adopts a separated core-board-and-baseboard design, with the core board connected to the baseboard via a BTB connector. The baseboard integrates serial ports, USB interfaces, Wi-Fi, Bluetooth, audio, Gigabit Ethernet, MIPI CSI ×2, MIPI DSI, 4G/5G module interfaces, CAN FD, RS485, RTC, and more.

It features powerful edge AI computing capabilities, supporting mixed INT8/INT16 precision operations. It covers core vision task networks such as object detection, image segmentation, and image classification. It also integrates advanced voice processing, including high-naturalness text-to-speech (TTS) and accurate automatic speech recognition (ASR), together with a full-process OCR text detection and recognition solution. The system is deeply compatible with mainstream deep learning frameworks including PyTorch, TensorFlow, and ONNX.

It supports H.265/H.264 hardware decoding up to 4K (3840×2160) @30 fps. It supports HEVC (H.265) Main Profile Level 5.0 High Tier, H.264 High Profile Level 5.0, and JPEG Baseline up to 12M @30 fps. It supports hardware encoding for mainstream resolutions including 4K (3840×2160) @30 fps and 1080P @30 fps.

It can be applied in intelligent security, smart door locks, industrial vision, smart home, intelligent automotive, robotics, and other fields.

The core board leads out **116 GPIO pins** (which can be multiplexed for other functions) and **60 functional pins** (for MIPI camera, MIPI display, USB, ADC, etc.).

Note: For bulk procurement of core boards, if your enterprise client is responsible for designing the matching baseboard, we sincerely invite you to initiate the cooperation process through the business customer service channel of our official flagship store.

For qualified enterprise customers, we will set up a **dedicated technical support group** to provide efficient and personalized service. This group will serve as a platform for rapid response and in-depth communication, fully accelerating the secondary development process of enterprise customers, ensuring smooth project progress, and helping your products quickly launch and gain a first-mover advantage in the market.

1.2 Core Board Pin Order and Exported Interface Signals

The ATK-CLRV1126B core board uses a board-to-board (BTB) connector. The pin information is divided into two parts: Part A and Part B. For details, please refer to the schematic diagram of the ATK-DLRV1126B baseboard.

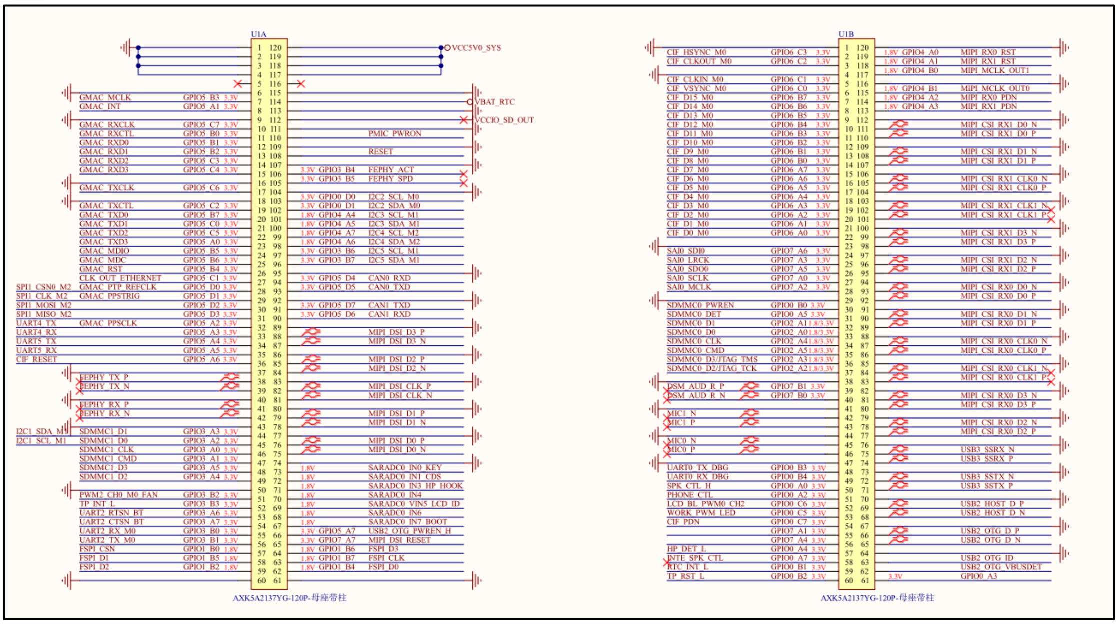

Figure 1.2-1 Core Board Pin Diagram

1.3 Core Board Pin Multiplexing Resources

The core board leads out all I/O pins of the processor. Users can design their own baseboard according to their needs to utilize the I/O resources on the core board and multiplex the I/O pins into the required functions.

According to the peripheral functions, the maximum number of available single peripherals that can be multiplexed on the ATK-CLRV1126B core board is listed here. The specific configuration can be combined with the chip datasheet. The following references are from the RV1126B chip datasheet

(Maximum number of single peripherals: refers to the maximum quantity of a single peripheral that the core board can use without employing other peripherals).

Some functions may be subject to third-party licensing requirements. Please refer to the Rockchip manual for details.

Pin Peripheral Functions	Maximum multiplexing quantity for a single peripheral	Note
MIPI Display Interface	1	4-lane MIPI DSI: supports up to 1920×1080@60fps
MIPI Camera Interface	2	4-lane MIPI CSI: transmission rate of 2.5 Gbps per lane
PWM	4	
SAI	3	
Ethernet	1	10/100M Ethernet controller Supports 10/100/1000Mbps via RGMII interface
USB 2.0	1	Can be expanded via USB HUB
USB3.0	1	Can be expanded via USB HUB
SPI	2	Supports master mode and serial slave mode.

I2C	6	Supports 7-bit and 10-bit address modes.
UART	8	There are 8 native channels, and uart0 is used as the debug serial port of the core board.
CAN	2	
ADC	3	There are 3 native SARADC channels.

*Note: The above peripheral data is based on the RV1126B chip datasheet. Whether peripheral functions conflict must be evaluated according to the actual project. You can combine the *RV1126B Core Board Pin Multiplexing Table* to configure the peripheral function combinations for specific core board pins.

Chapter 2. Notes on Core Board Usage

Before using the ATK-CLRV1126B Core Board for function verification and project development, users should pay attention to the following items:

Precautions

- The ATK-CLRV1126B Core Board **cannot work independently**; it must be used with a matching baseboard.
- Users are recommended to purchase the ATK-DLRV1126B Development Kit, which includes a baseboard and the ATK-CLRV1126B Core Board.
- All core boards are pre-programmed with the factory system firmware and functionally tested before shipment.
- After receiving the development kit, users can perform functional and hardware tests by following the document

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located in the User Manual section of the cloud disk resources, to ensure the core board operates properly.

- Before connecting the ATK-CLRV1126B Core Board to your custom baseboard, you must test the reliability of the baseboard. Ensure that the power supply to the core board is stable, there are no short circuits or cold solder joints on the baseboard, and the voltage levels of the baseboard peripheral pins comply with the relevant power domains of the ATK-CLRV1126B Core Board. Make sure the circuit is normal and the soldering quality is reliable. Improperly connecting the core board directly to your baseboard may result in permanent damage to the core board.
- DO NOT plug or unplug the core board and peripheral modules while power is on!
- Before using the product, carefully read this manual and related development documents, and observe the platform application notes.
- Follow all instructions and warning information marked on the product.
- Use this product in a cool, dry and clean environment.
- Keep this product dry. If splashed or soaked by any liquid, disconnect power immediately and let it dry completely.
- Do not clean this product with organic solvents or corrosive liquids.
- Do not use or store this product in dusty or dirty environments.
- If the product will not be used for a long time, pack it properly to prevent moisture and dust.
- Ensure sufficient ventilation and heat dissipation during use to avoid component damage caused by overheating.
- Do not use the product in environments with rapid temperature changes to avoid component damage due to condensation.
- Do not handle the product roughly; dropping, knocking or violent shaking may damage circuits and components.
- Take strict anti-static precautions when using this product.
- Do not repair or disassemble the product by yourself. If a fault occurs, contact our company for maintenance in a timely manner.

- Unauthorized modification or use of unapproved accessories may damage the product. Damage caused by such actions will not be covered under warranty.

Chapter 3. Minimum System Design Recommendations

3.1 System Power Supply Design

The onboard power supply circuit diagram of the ATK-DLRV1126B development board is shown below:

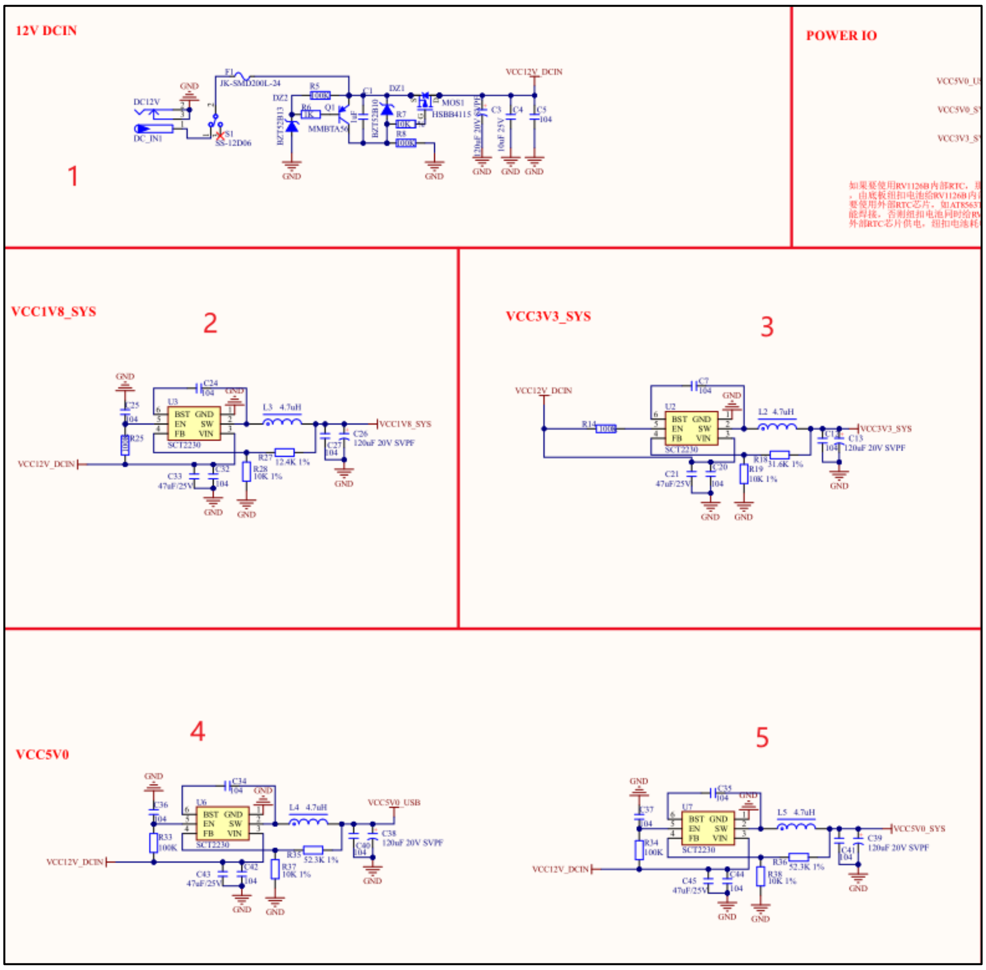

Figure 3.1-1 Power Supply Section of the Development Board

As can be seen from the diagram above, during the power supply process:

① The development board is powered by a DC 12V/1A power adapter, with the main power input through the DC_IN interface.

② One power path is converted by the SCT2230 chip into **VCC1V8_SYS** for the baseboard peripherals.

③ One power path is converted by the SCT2230 chip into **VCC3V3_SYS** for the baseboard peripherals.

④ One power path is converted by the SCT2230 chip into **VCC5V0_USB** for USB usage.

⑤ One power path is converted by the SCT2230 chip into **VCC5V0_SYS** for powering the core board and baseboard peripherals.

When designing the minimum system board for the core board, **VCC5V0_SYS** at position ⑤ must be supplied to the core board.

Other power supplies are determined by the actual power requirements of the baseboard peripherals.

You may refer to the power supply design of the corresponding peripherals on the ATK-DLRV1126B baseboard.

3.2 Development Board Reset Circuit Design

The RESET pin led out from the core board is Pin 108 of the core board Part A, which is a reset signal pin. It is active low and requires a minimum reset time of at least 4 µs.

To improve anti-interference ability and prevent system reset caused by false triggering, it is recommended to parallel a 100 nF capacitor (C90 in the schematic diagram) and place it close to the core board pin.

If the reset signal is not used, the RESET pin should be left floating.

The reset signal should be routed away from clock signals, and the RESET pin must not be connected to ground.

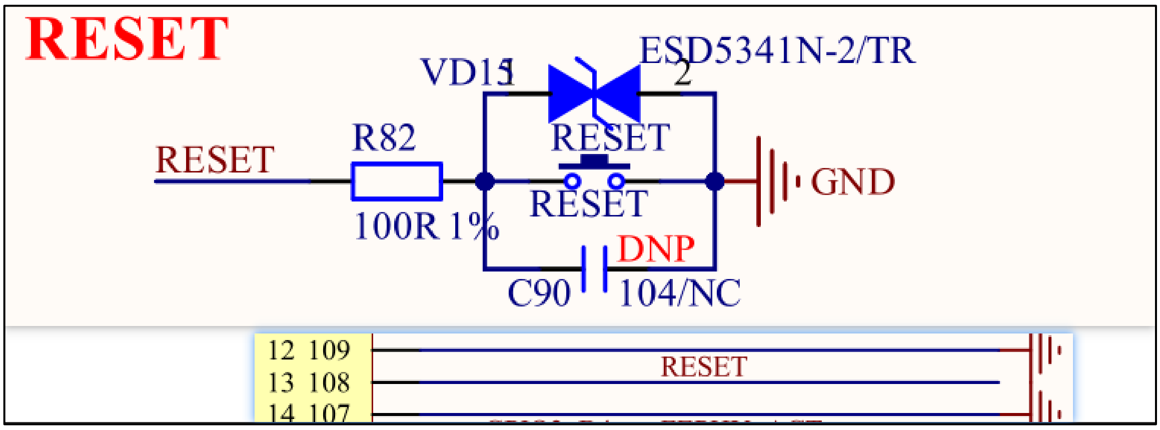

Figure 3.2-1 Reset Button

3.3 Development Board Function Button Design

There are 7 function buttons on the development board, namely RST, PWRON, UPDATE, ESC, V+, V-, and MENU. You can select and keep them according to your project requirements.

The RESET button is used to reset the entire development board, corresponding to the reset signal pin Pin108 of Part A on the core board. It is recommended to keep this button.

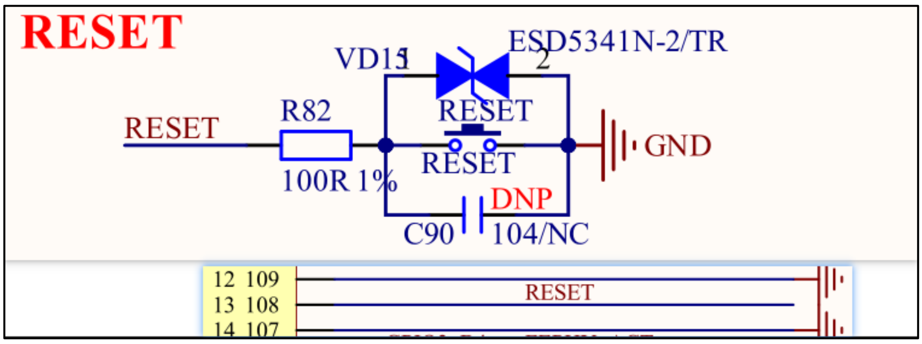

Figure 3.3-1 Reset Button

The PWRON button is used to control the power on/off of the RK801 and can be used for low-power switching. It corresponds to pin 110 of Part A on the core board. You may choose to keep it according to project requirements.

It is active low and should be left floating if not used.

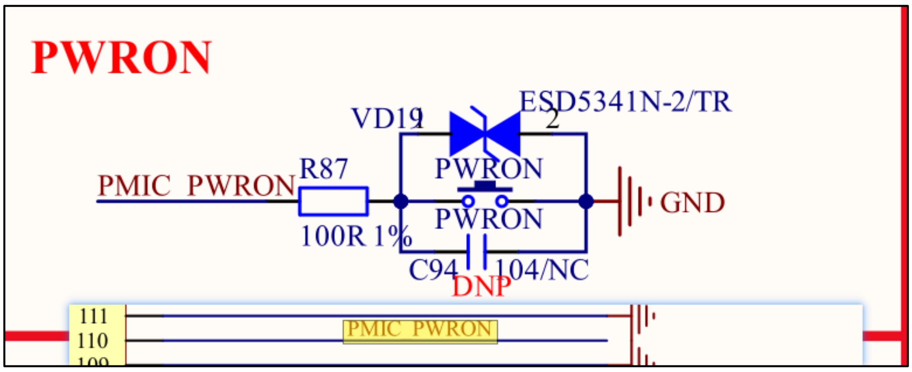

Figure 3.3-2 PWRON Button

The UPDATE button is used for firmware upgrade of the development board. Hold down the UPDATE button and press the RESET button to reset; the system will enter MaskRom firmware burning mode. It is recommended to keep this button. It corresponds to pin 67 of Part A on the core board.

It is controlled by SARADC0_IN7 and is valid when the sampled voltage reaches 0V. The pin should be left floating when not in use.

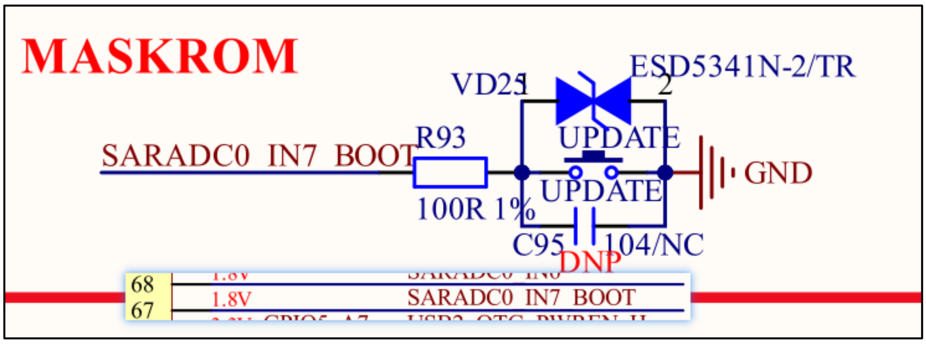

Figure 3.3-3 MASKROM Button +

The MENU, ESC, V+ and V- buttons are main menu function buttons. You may keep them according to project requirements.

All these input buttons are controlled by **SARADC_VIN0** on **pin73** of Part A of the core board.

They are activated when the corresponding sampling voltage value is reached.

If these functions are not used, the pin shall be left floating.

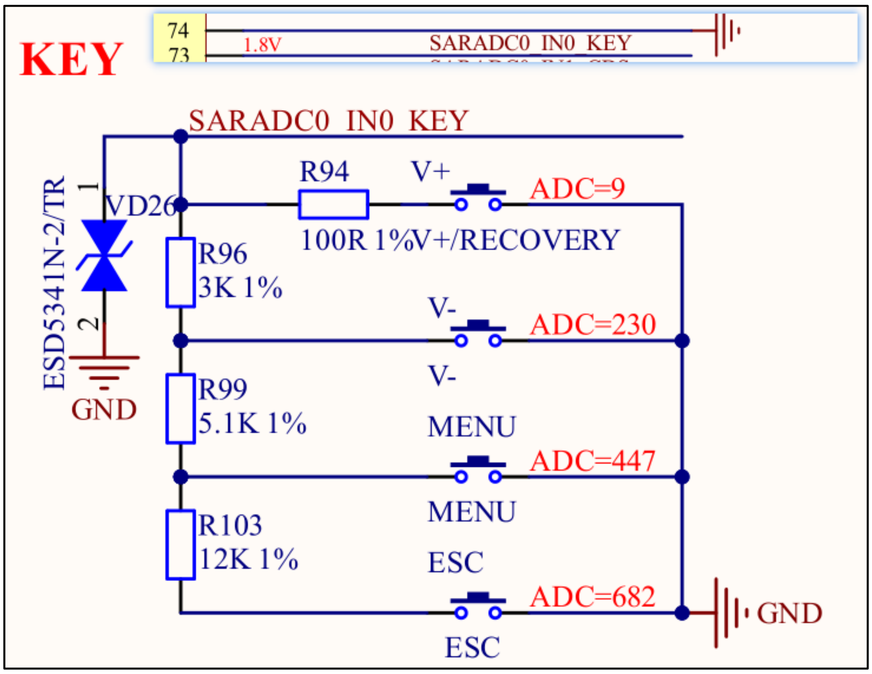

Figure 3.3-4 Function Buttons

3.4 UART Debug Interface

The UART interface of the development board uses a Type-C connector, which connects to a computer via a USB-to-Type-C cable. The onboard CH343G chip converts the computer USB signal into the development board UART signal to realize serial communication.

It is recommended to keep the default UART debug interface of the development board for printing system information. UART0 is used by default and cannot be replaced with other UARTs. The development board UART connector uses a 16-pin Type-C interface.

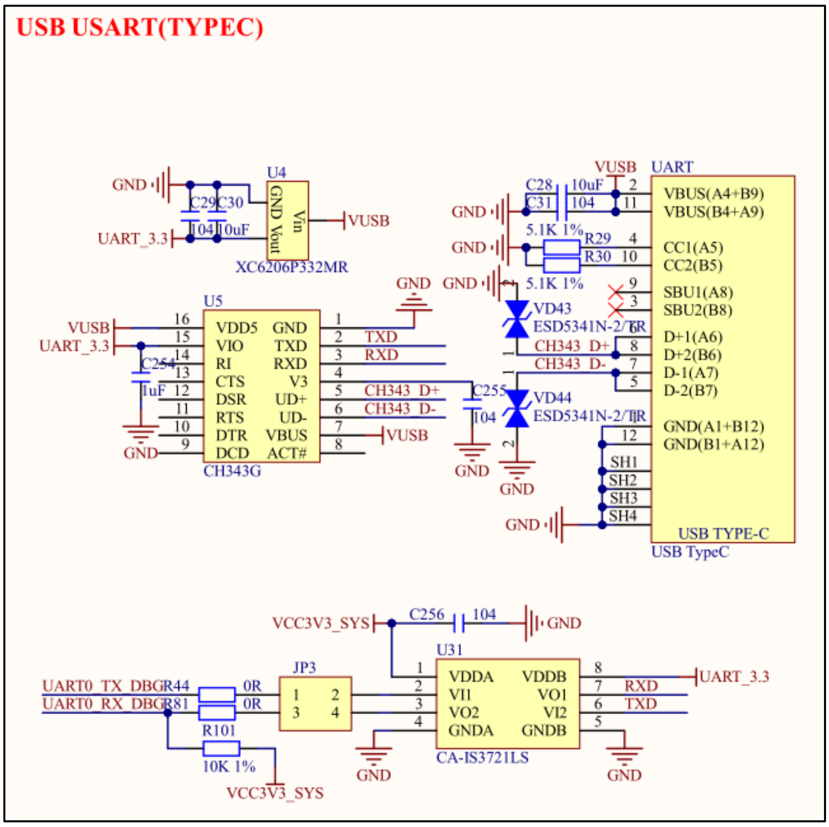

Figure 3.4-1 UART Circuit Design

USB TTL (USB Type-C) Pin Description:

Pin Number	Pin Name	Type	Description
2、11	VBUS	Power	VUSB power input, 5V input after filtering.
4、10	CC1\2	CC Signal	As a slave device, the development board requires a 5.1K pull-down resistor to GND.
3、9	SBU2\SUB1	No Connection	Auxiliary signal, not required here, leave floating.
5~8	D1\2±	USB Signal	USB 2.0 data lines.
1、12	GND	Power	Ground.
SH1~4	SH1~4	Power	Ground, for connector fixation.

CH343G Pin Description:

Pin Number	Pin Name	Type	Description
1	GND	Power	Ground
2	TXD	Output	Serial data output
3	RXD	Input	Serial data input
4	GND	Power	Ground, external 104 capacitor required
5	CH343 D+	USB Signal	Directly connected to D+ data line of USB bus
6	CH343 D-	USB Signal	Directly connected to D- data line of USB bus
8~14		No Connection	

15	UART_3.3	Power	Positive power input pin, an external 1μF capacitor is required.
7、16	VSUB	Power	Positive power input pin.

On the development board, jumper cap JP3 is used to connect UART0 and CH343G.

UART0_TX_DBG connects to RXD of CH343G, and UART2_RX_DBG connects to TXD of CH343G.

If a jumper cap is not required, direct connection is acceptable.

UART2_RX_DBG and UART2_TX_DBG correspond to pin48 and pin47 of Part B on the core board respectively.

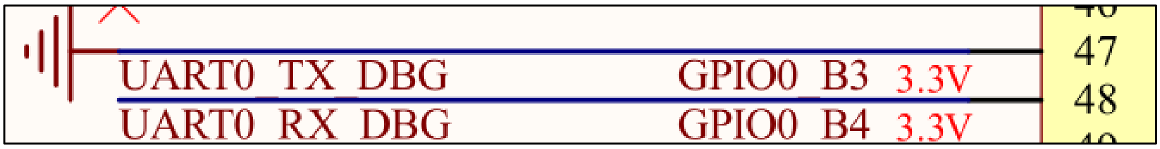

Figure 3.4-2 Core Board Pins